

Experiment No. 6

Aim: To measure inductance of a given coil by Anderson's bridge method.

Apparatus:

Inductor, Standard Capacitor, Resistors (fixed resistance's and variable pots as given circuit), Signal generator, Head phones and Connecting terminals.

Theory:

Anderson's bridge is the most accurate bridge used for the measurement of self-inductance over a wide range of values, from micro-Henry's to several Henry's. In this method the unknown self-inductance is measured in terms of known capacitances and resistances, by comparison. It is a modified of Maxwell's L-C bridge. In this bridge, double balance is obtained by the variation of resistances only, the value of capacitance being fixed. This method is high accurate for determining inductance. Q factor cannot be measured by this method since we cannot bring the bridge to resonance condition.

Formula:

Inductance of given coil

$$L = C * [R_5(R_1 + R_2) + R_3 * R_4] \text{mH}$$

Where,

C =Capacitance of the standard capacitor (μF)

R_2, R_3, R_4 = known, fixed and non-inductive resistances ($\text{K}\Omega$)

R_1, R_5 = Variable resistances ($\text{K}\Omega$)

Procedure:

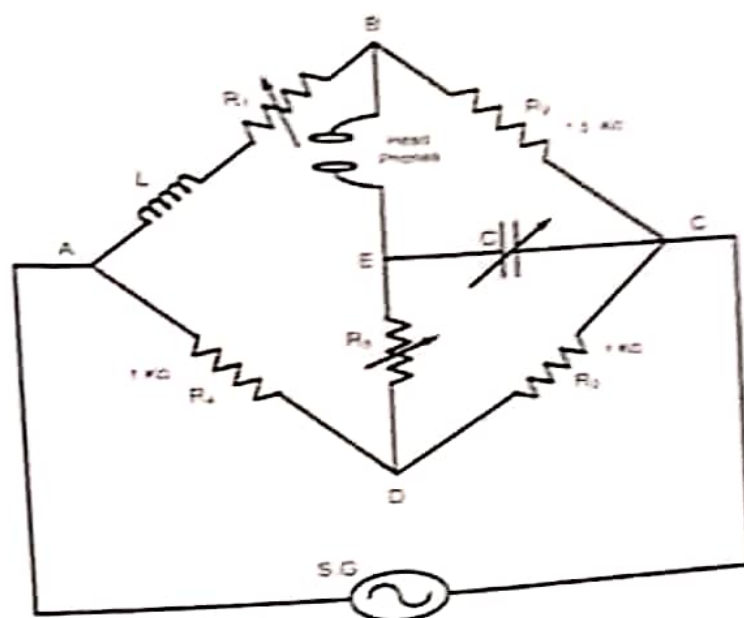
This circuit diagram of the bridge is as shown in figure. The coil whose self-inductance is to be determined, is connected in the arm AB, in series with a variable non-inductive resistor R_1 . Arms BC and DA contain fixed and non-inductive resistors R_2 , R_3 and R_4 respectively. Another non-inductive resistor R_5 is connected in series with a standard capacitor C and this combination is put in parallel with the arm CD. The head-phones are connected between B and E. The signal generator is connected between A and C junctions.

Select one capacitor and one inductor and connect them in appropriate places using patch cords. The signal generator frequency is adjusted to audible range. A perfect balance is obtained by adjusting R_1 and R_5 alternatively till the head-phones indicate a minimum sound. The values of R_1 and R_5 are measured with a multi-meter (While measuring the R_1 and R_5 values, they should be in open circuit). In the balance condition the self-inductance of the coil is calculated by using the above formula. The experiment is repeated with different values of capacitances.

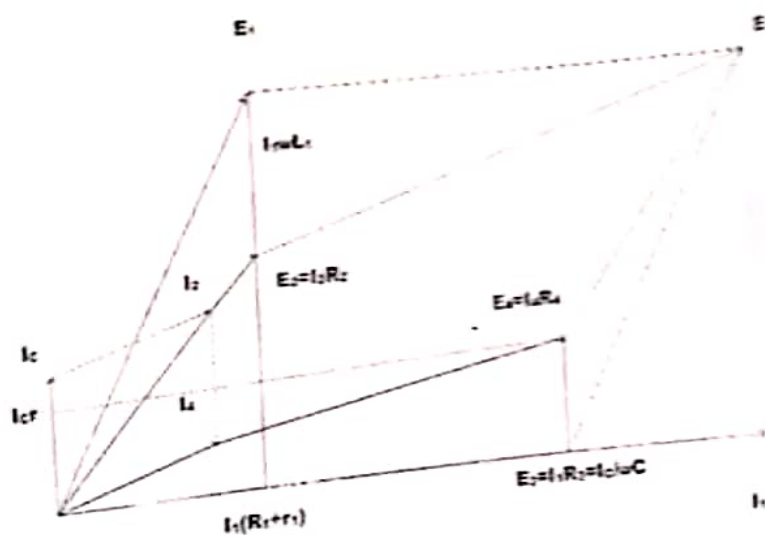
Observation Table:

S.No	Capacitor C (μF)	Resistance R_1 (Ω)	Resistance R_5 (Ω)	Calculated values of $L = C * [R_5(R_1 + R_2) + R_3 * R_4]$ mH	Standard value of L (mH)
1.					
2.					
3.					
4.					
5.					
6.					

Circuit Diagram:



Anderson's bridge



Phasor diagram

Precautions:

1. The product (CR_2R_4) must always be less than L .
2. R , and R_s are adjusted until a minimum sound is heard in head-phones.

Result:

The inductance of given coil is _____ mH

Questions: